

Noise-Aware ANNNI Phase Diagnostics

Low-CNOT preparation, error mitigation, and their limits

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Model and approach

We study the periodic axial next-nearest-neighbor Ising chain $H = -\sum ZZ(nn) + \kappa \sum ZZ(nnn) - h \sum X$, $J_1=1$. Competing interactions complicate state preparation and phase identification at low field. We evaluate state accuracy and observable errors separately from the evidence for phase boundaries. The $N=8$ circuits cover 420 positive-field coordinates. $N=12$ circuits and larger matrix-product-state (MPS) calculations with open boundaries extend the study across system sizes and boundary conditions.

Preparation, noise and diagnostics

B0 is a fixed six-layer Hamiltonian variational ansatz. B3 adaptively grows energy-optimized Pauli rotations from multiple reference states and selects a circuit using a frozen energy/resource rule. After every executed CNOT, its target receives $D_p(\rho) = (1-p)\rho + p/3(X\rho X + Y\rho Y + Z\rho Z)$. This includes CNOTs used for reference preparation and folding. The explicit gate implementation preserves the recorded connectivity and wire-0-MSB convention. We measure the full $C(r)$, all discrete structure factors and M_x . Frozen D3 uses PCA and clustering to classify resemblance to reference patterns; it supplies a diagnostic label rather than independent phase truth.

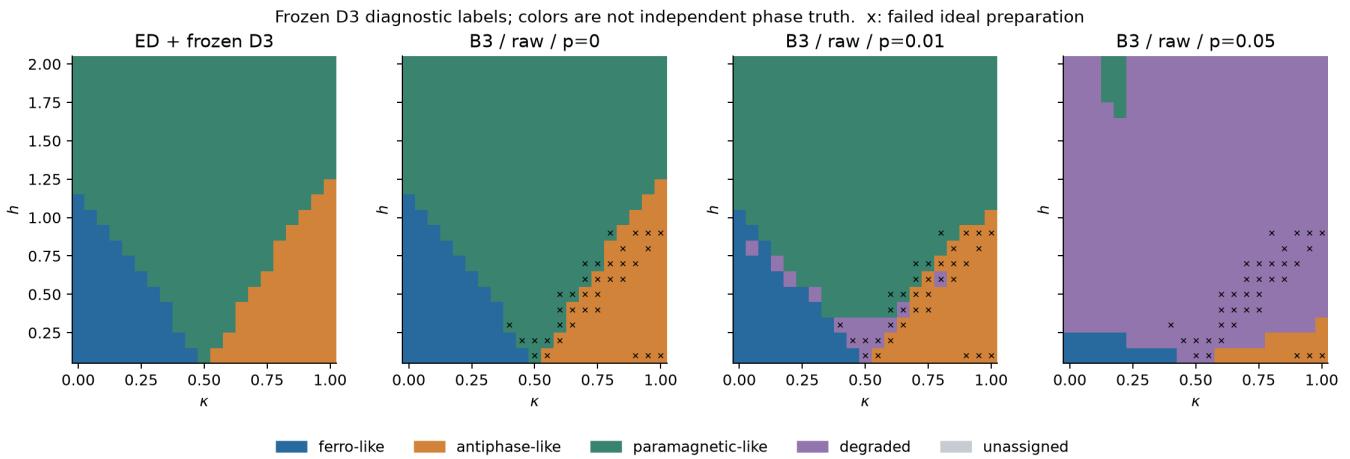
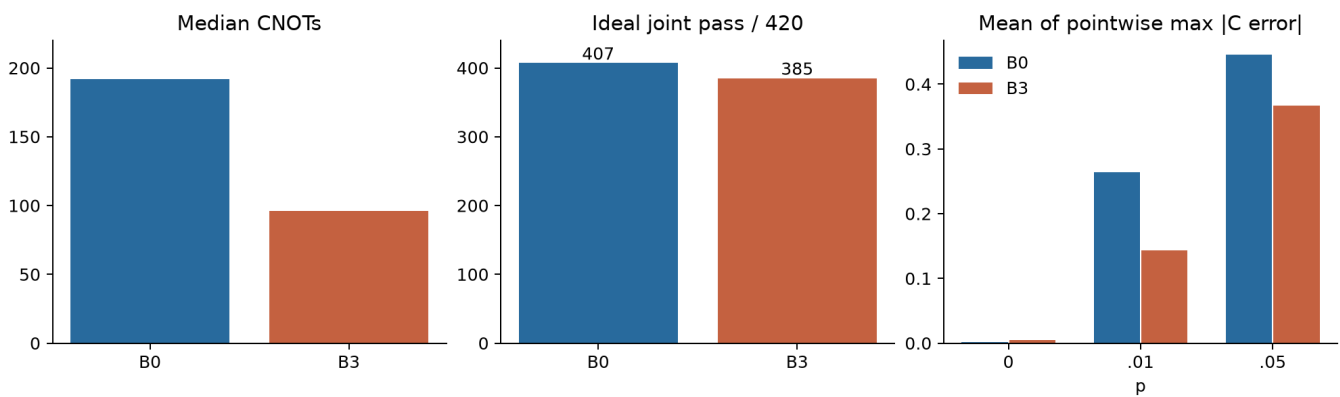


Figure 1. N8 PBC, $h>0$. The ED reference and raw B3 results at three noise levels use frozen D3 and the same color scale. Colors show phase-like estimates; crosses mark failed preparations. The notebook shows reference coverage separately.

Fewer CNOTs, lower preparation coverage

Replacing B0 with B3 reduces median CNOTs from 192 to 96, while ideal joint preparation passes fall from 407/420 to 385/420. At $p=.01$, the mean pointwise maximum correlation error falls from 0.2638 to 0.1432, with improvement at 402/420 coordinates. These values measure correlation reconstruction, not phase-classification accuracy. A joint pass requires $\delta e \leq .001$ per spin, $C/SF/M_x$ errors $\leq .02$ and squared ED fidelity $\geq .99$.



Comparison at equal CNOT-shots

We compare raw estimates, quadratic zero-noise extrapolation (ZNE) and global-X symmetry verification (SV) on 45 fixed coordinates, with 32 measurement repeats. Each method has the same per-coordinate budget $G = \sum(\text{shots} \times \text{literal CNOTs})$. Raw and SV use 100,000 shots; ZNE uses 33,334 across folds 1/3/5, with weights 15/8, -5/4, 3/8. SV estimates $(O+OP)/(1+P)$ using all required signed joint settings. Matching G controls CNOT-weighted shot cost. Measurement settings and one-qubit costs remain outside this budget.

Observable errors and the zero-noise control

The ED-target MSE gives equal weight to the same 17 components: eight C, eight Fourier-related SF, and Mx. At $p=.01$, raw/ZNE/SV give 0.01754430/0.00162271/0.01235293; at $p=.05$ they give 0.10511371/0.07863611/0.10231166. ZNE lowers mean MSE by 90.75% and 25.19%, respectively, and improves the coordinate-mean error at all 45 points. At $p=0$, it worsens all 45. The reduction in bias comes with higher sampling MSE. Only six points have independent interior labels, so the 45-point result measures observable reconstruction rather than phase accuracy.

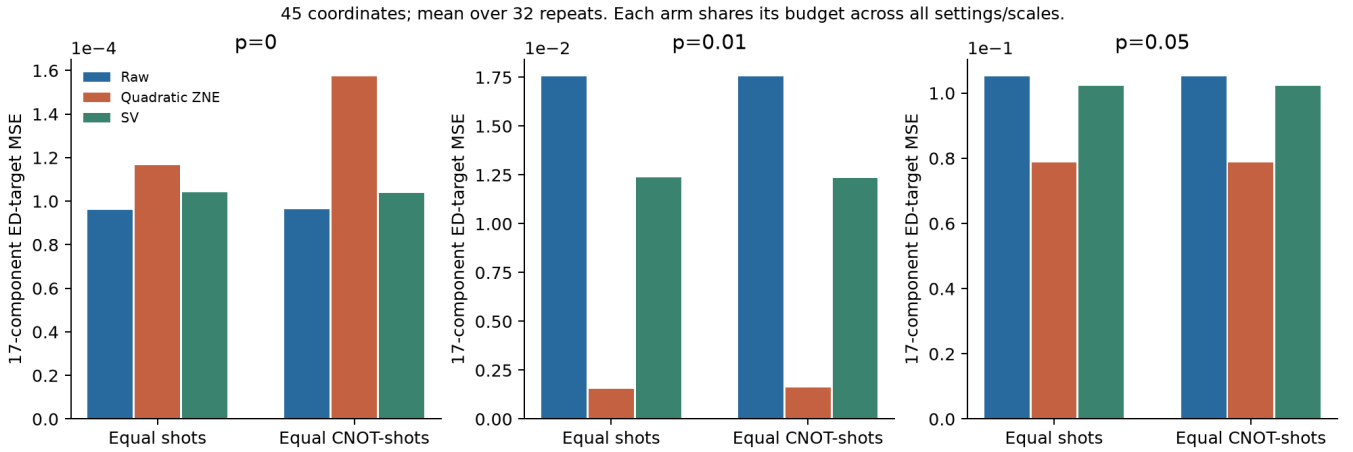


Figure 2. Results include all 45 coordinates. Each value averages 32 repeats, with the original equal weighting of related C/SF components. The panels compare experiments matched by shots and by CNOT-shots.

Response-feature matching

We execute all six 17-point windows at $\kappa=0, .3, .45, .5, .55, .8$ using identical grids and frozen matching rules. The .5 reference fails the frozen peak-selection criterion. The other five references describe finite-size RN features, not thermodynamic critical points. At $p=.05$ and 100k equal shots, raw estimates match 2.96875/5 window equivalents across repeats, compared with 2.875/5 for ZNE. The records retain endpoint domination, competing peaks, preparation errors and branch sensitivity. These results show why observable MSE and finite-size feature matching need separate evaluation.

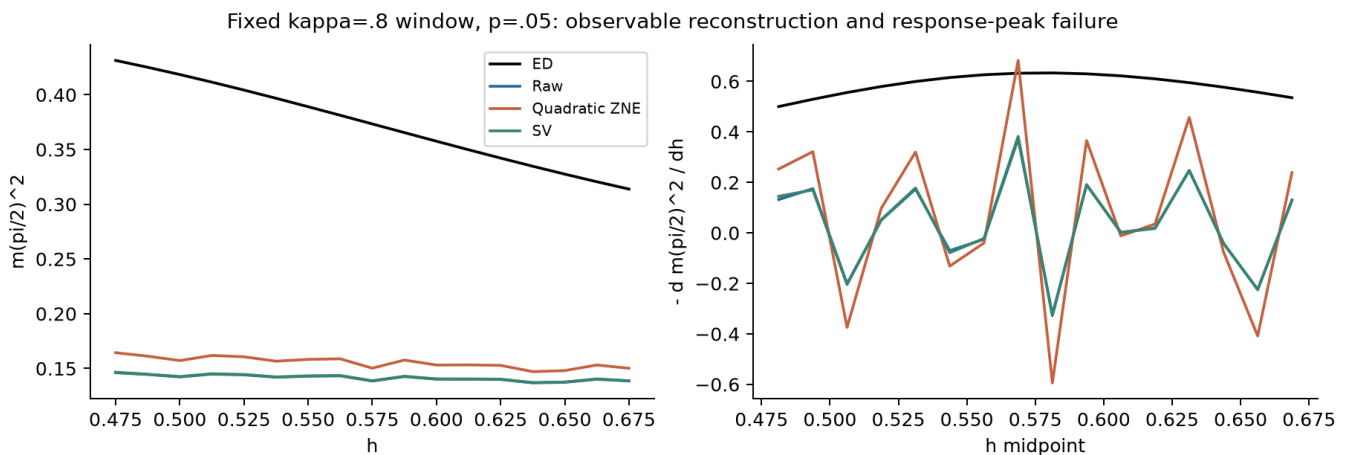


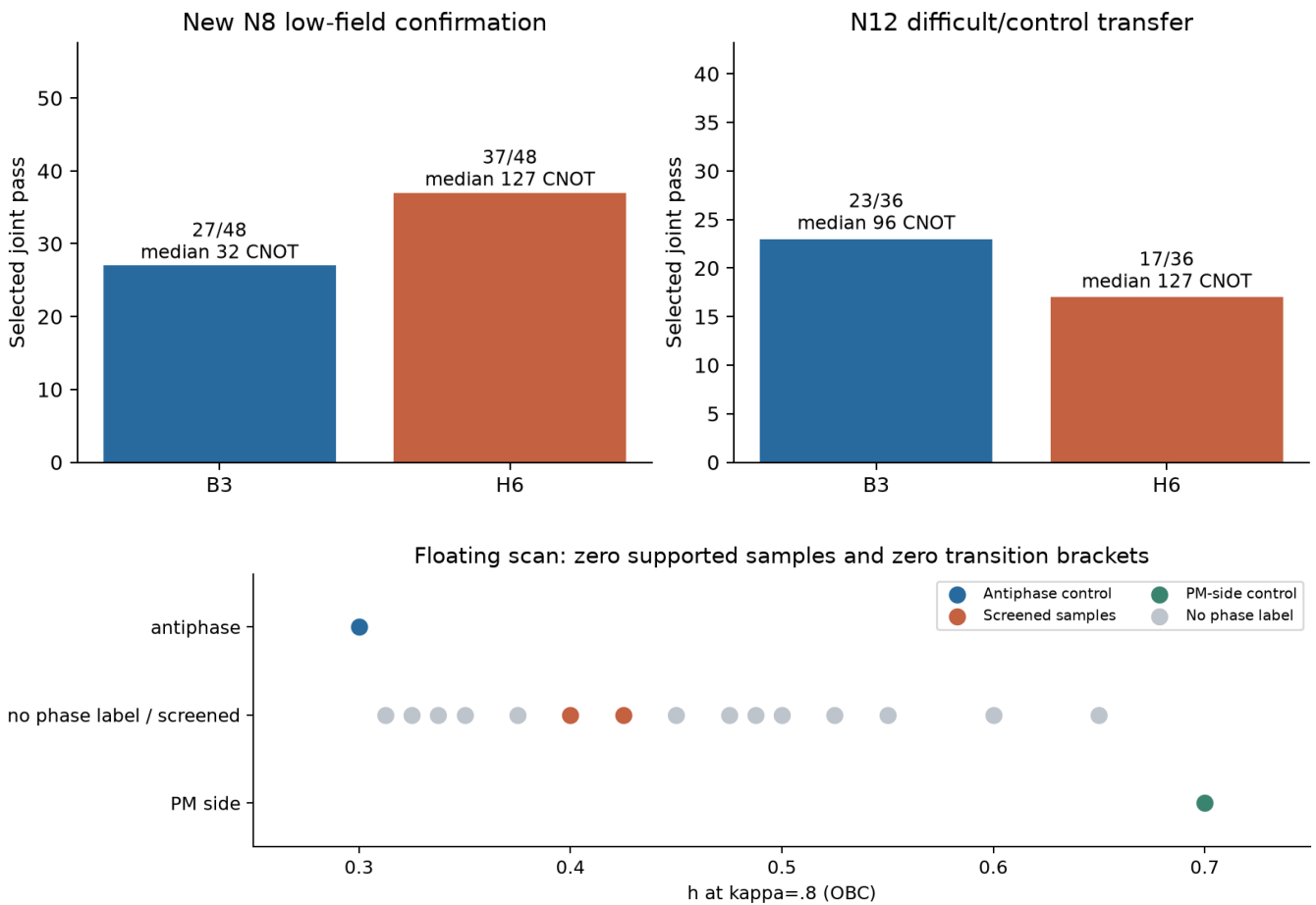
Figure 3. The prespecified $\kappa=.8$ window at $p=.05$ uses exact expectations to isolate noise and preparation effects. All endpoints and both competing response features are retained. Peaks are not selected afterward by proximity to the reference.

Low-field preparation and transfer

H6 combines physical and domain-wall candidates in a frozen union. On the N8 low-field cohort, candidate availability rises from 27/48 to 45/48 and selected joint passes from 27/48 to 37/48. This improvement costs more: median CNOTs rise from 32 to 127, and objective calls increase by 7.45x. On the N12 difficult/control transfer set, B3 passes 23/36 and H6 passes 17/36. H6 therefore improves the N8 low-field result at greater cost but reduces the pass count on this N12 cohort.

Floating-phase search and extensions

The controlled $\kappa=0.8$ OBC slice supports an antiphase sample at $h=0.3$ and a paramagnetic-side sample at $h=0.7$. At $h=0.425$, the size, entropy and Friedel diagnostics disagree. The final evidence contains zero supported floating samples and zero established transition brackets. Bond-dimension convergence checks numerical stability; phase identification depends on gap and correlation-length scaling. The $\chi=512$ run reached its time limit, which provides no basis for excluding a phase. The archived dynamics quantify Trotter and noise effects. The notebook also contains the multiple-detector and N12 experiments, all drawn from the frozen dataset.



Conclusions and reproduction

Lower-gate preparation and cost-controlled mitigation improve observable reconstruction in these experiments. Reliable phase-boundary inference remains a separate requirement. We distinguish R0 region interiors, RN same-size diagnostics and Rlarge evidence qualified by system size. The main figures and statistics rebuild from the bundled numerical data and joint counts. The package also includes lightweight explicit-gate and SDK checks, with separate results for V1 deterministic saved-data verification, V2 physical checks and V3 exact random replay under its specified environment contract.

References: Q-SITE 2026 Scientific Track, snapshot 57f9a537 (local upstream/); Tang et al., qubit-ADAPT-VQE, arXiv:1911.10205; ANNNI tensor-network study, arXiv:2402.11022; floating-phase DMRG, arXiv:cond-mat/0702676; symmetry verification, arXiv:1807.10050. Published XX/Z conventions are related to ZZ/X by local Hadamards; boundaries and sizes are not interchangeable.